

Particle Identification in Highly Granular Calorimeters With Deep Sets

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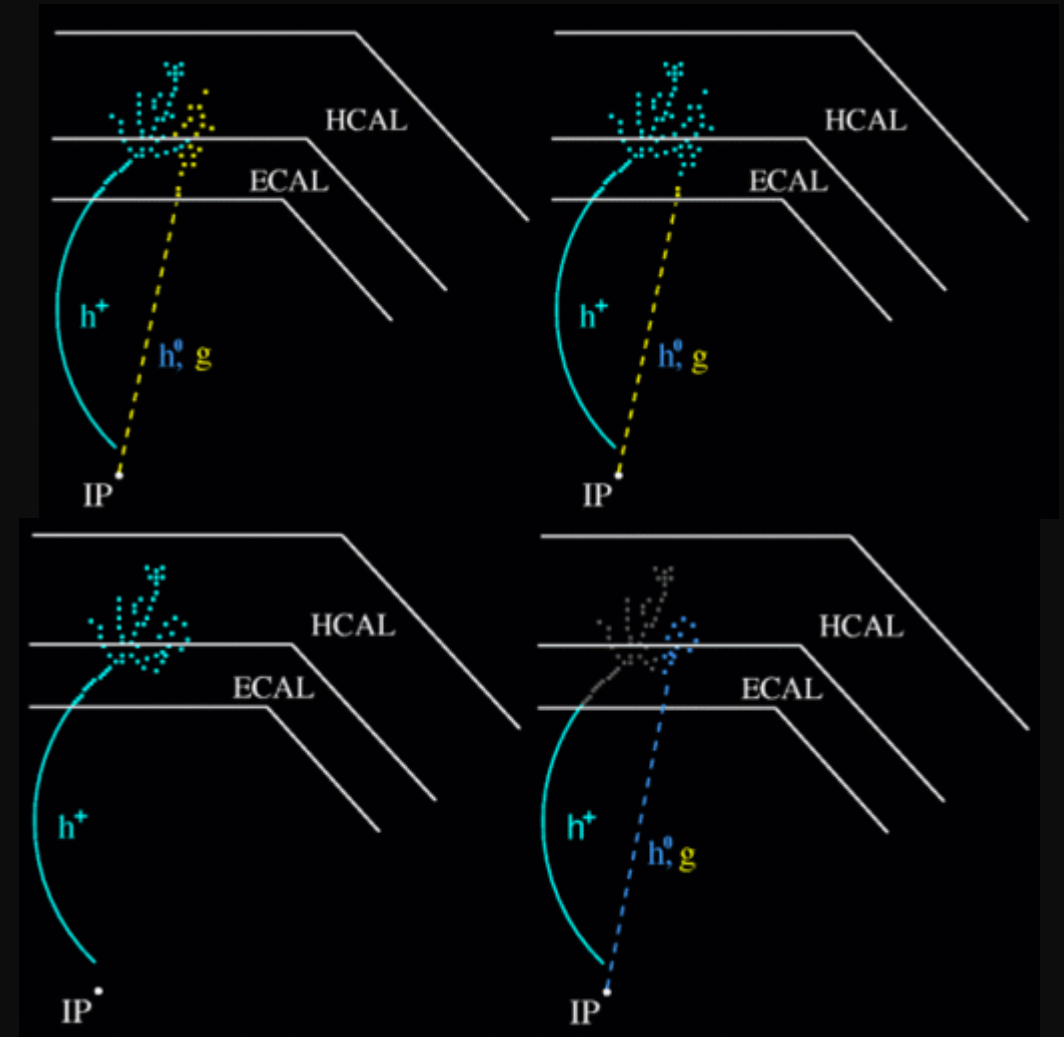
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What is a calorimeter ?

- Calorimeters are used to measure the energy of particles :
 - 1- Neutral particles
 - 2- High energy charged particles
- Primary tool for jet energy measurement
 - Used to measure the energy of particle, crucial for reconstructing the total jet energy ($E_{\text{JET}} = E_{\text{ECAL}} + E_{\text{HCAL}}$).
 - “Typical” jet:
 - ~62% charged particles (mainly hadrons)
 - ~27% photons
 - ~10% neutral hadrons
 - ~1% neutrinos [\(Source\)](#)



Why do we need high granularity calorimeter?

- FCC will study physics at energy scale of “**tens of TeV**”.
- Jets will be highly boosted at such energy scale.
- High jet energy resolution is essential, also for Higgs factories >90% of events have >2 jets. So having excellent JER will help study particles and their interactions
- High granularity improves pile-up robustness
- Role evolved after following developments:
 - **Particle Flow Algorithm**: Reconstructs individual particle using the most precise subdetector.
 - **Boosted Jet Tagging**: It requires fine substructure resolution in jets.
 - **Time resolution?**

Objective

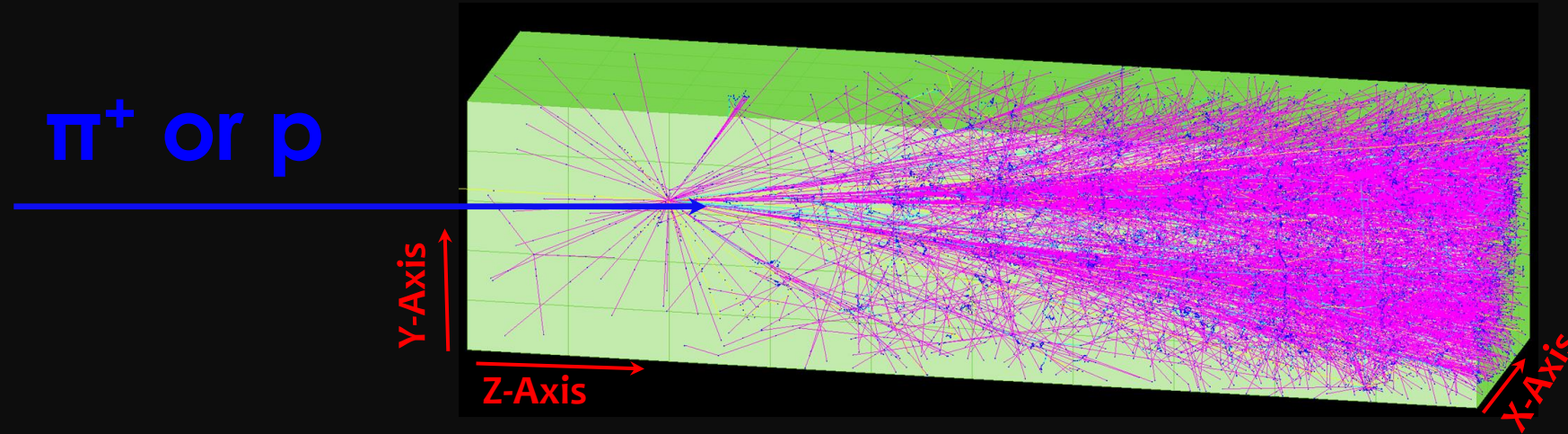
- To understand the maximum extractable information from a calorimeter.
- A few assumptions are made for this study:
 - Perfect time resolution inside the calorimeter.
 - Perfect spatial resolution.
 - Arbitrary high granularity.
- GEANT4 is used for simulating the homogeneous calorimeter:
 - State of art scintillating material: **Lead Tungstate (PbWO₄)**.
 - Segmentation: 100 x 100 x 200 cells
 - Cell size: 3 x 3 x 6 mm³
- Granularity varied to study the effect of loss of information.
 - i.e. Z-axis: 100 x 100 x **200** → 100 x 100 x **100** → 100 x 100 x **50**



A 3 x 3 x 15 cm³ Lead Tungstate crystal and it costs 220 USD.

Data Generation

- **GEANT4 Particle Gun** : Proton and Pion are generated at energies: 25 GeV, 50 GeV and 100 GeV.

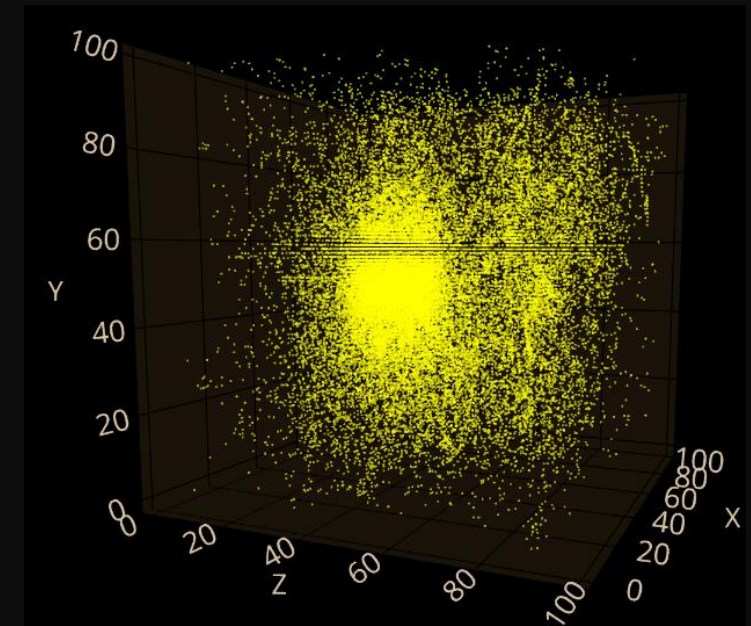
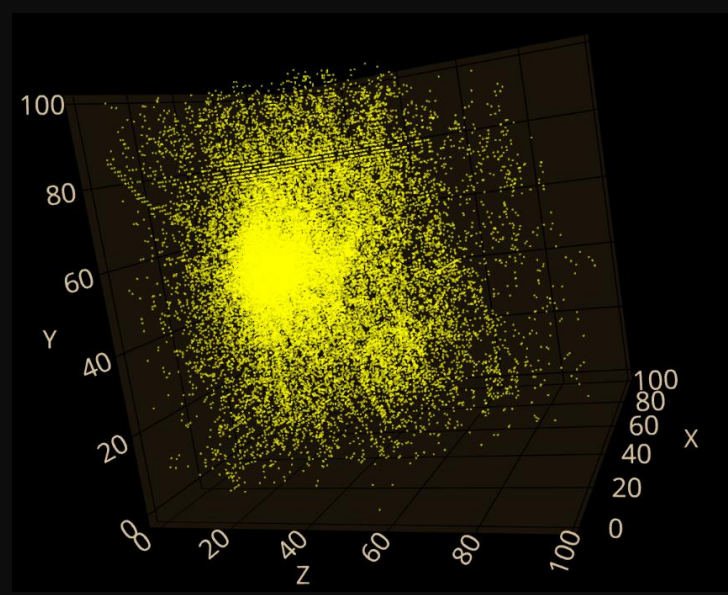
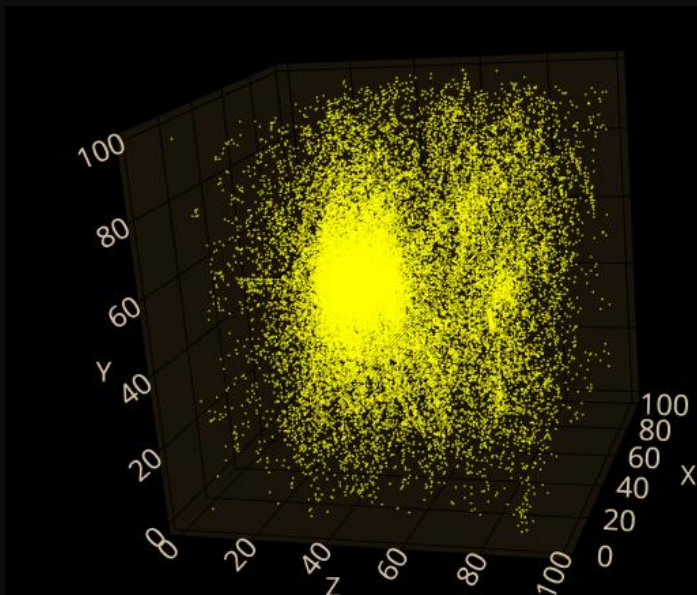


- Hadron showers are stored in ROOT files.
- Each file contains 100K events per hadron.
- Each shower will be represented as a Point Cloud.
- Both temporal and spatial features are included in the Point Cloud.

Data Generation

- Point Cloud: A point cloud is an unordered collection of data points in 3D space.
- Hadron shower as Point Cloud:
 - Cell Coordinates (X,Y,Z)
 - Cell Energy Deposited
 - Cell energy weighted time average
- Point Clouds are highly efficient to represent the showers in granular calorimeter.
- Following are few event displays for p and π^+ :

	Cell_x (idx)	Cell_y (idx)	Cell_z (idx)	Cell_E (MeV)	Cell_T (ns)
	50	49	0	6.7661	1.3549
	50	49	1	6.9507	1.3748
	.	.	.	.	.
	.	.	.	.	.
	94	87	157	0.0018	463.7056



High Level Features

- Two kinds of features were developed:
 - Global Features:** Describe overall shower characteristics independent of calorimeter granularity.
 - Example: Energy deposited in full calorimeter.
 - Local Features:** Capture shower properties using low-level information from individual cells.
 - Example: Radius of hadronic shower.

Open Access Article

Hadron Identification Prospects with Granular Calorimeters

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Abstract

In this work we consider the problem of determining the identity of hadrons at high energies based on the topology of their energy depositions in dense matter, along with the time of the interactions. Using GEANT4 simulations of a homogeneous lead tungstate calorimeter with high transverse and longitudinal segmentation, we investigated the discrimination of protons, positive pions, and positive kaons at 100 GeV. The analysis focuses on the impact of calorimeter granularity by progressively merging detector cells and extracting features like energy deposition patterns and timing information. Two machine learning approaches, XGBoost and fully connected deep neural networks, were employed to assess the classification performance across particle pairs. The results indicate that fine segmentation improves particle discrimination, with higher granularity yielding more detailed characterization of energy showers. Additionally, the results highlight the importance of shower radius, energy fractions, and timing variables in distinguishing particle types. The XGBoost model demonstrated computational efficiency and interpretability advantages over deep learning for tabular data structures, while achieving similar classification performance. This motivates further work required to combine high- and low-level feature analysis, e.g., using convolutional and graph-based neural networks, and extending the study to a broader range of particle energies and types.

Keywords: particle detectors; calorimetry; particle identification; physics; machine learning

High Level Features

Local Variables

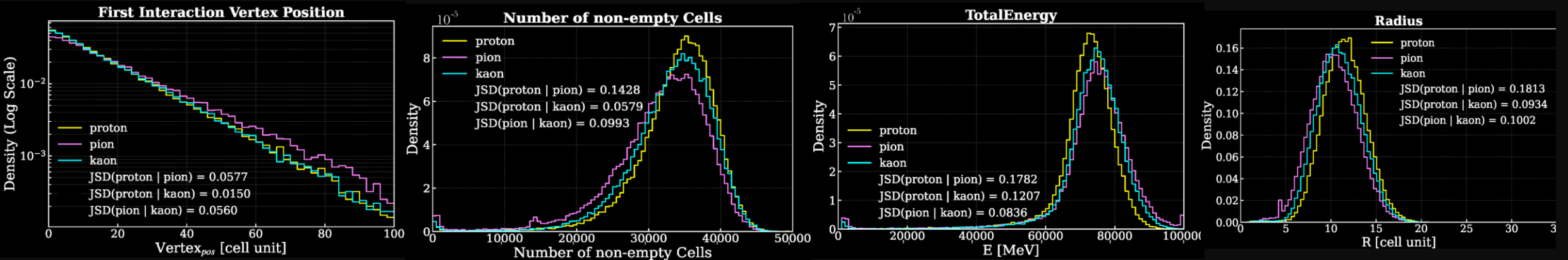
1. First nuclear interaction vertex position
2. First interaction vertex time
3. Speed
4. Δt
5. Fraction of energy after first vertex
6. Number of non-empty cells before vertex
7. Number of non-empty cells
8. Maximum cell energy
9. Second maximum cell energy
10. Total energy near vertex
11. Fraction of energy near vertex
12. Max energy near vertex
13. Energy variance near vertex
14. Distance to max energy cell
15. Distance between max and second-max cells
16. Energy near shower peak
17. Fraction of energy near shower peak
18. Left-right energy asymmetry (ELR)
19. Energy ratio (R_{cell_E})
20. Energy delta (Δ_{cell_E})
21. Energy fraction (FE)
22. Shower radius (R)
23. Shower radius σ (σR)
24. Shower length (L)
25. Shower length σ (σL)
26. Weighted radius (R_w)
27. Weighted radius σ ($\sigma_w R$)
28. Weighted shower length (L_w)
29. Weighted shower length σ ($\sigma_w L$)
30. Asymmetry (A)
31. Weighted asymmetry (A_w)

Global Variables

1. Total energy deposited
2. Calorimeter characteristic time
3. Time of flight (ToF)

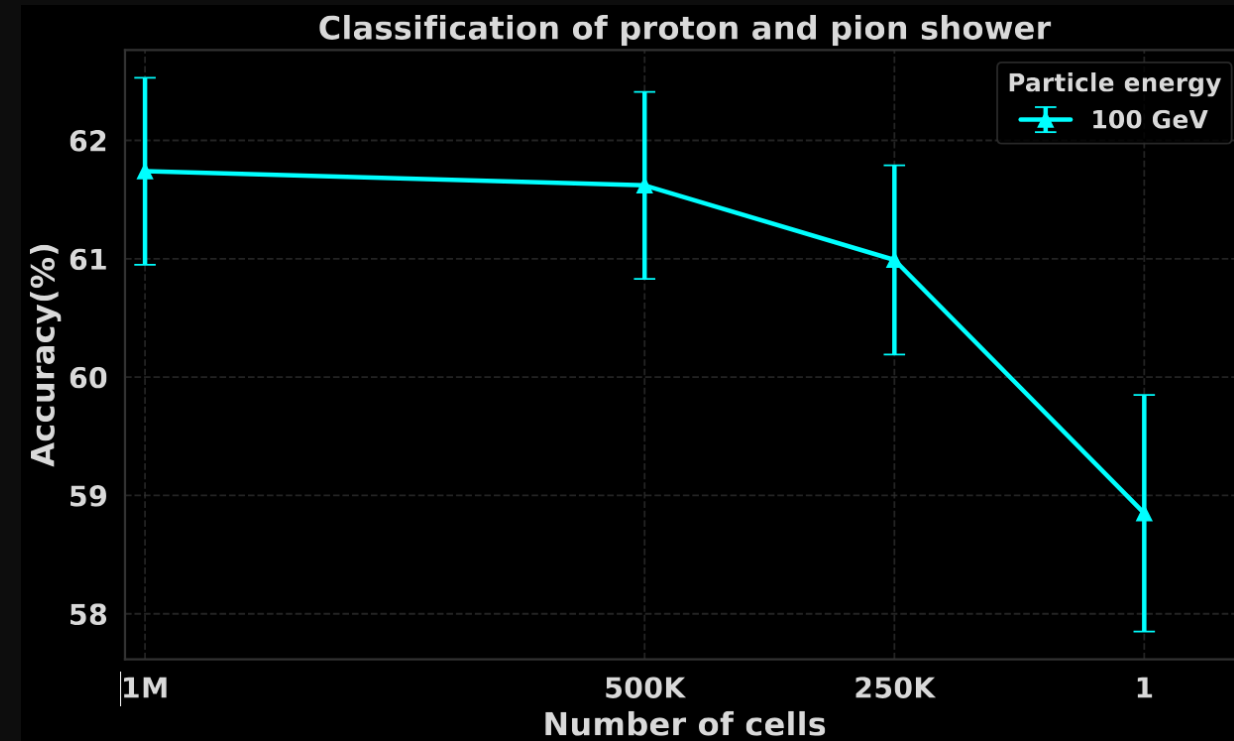
High Level Features

- **First Nuclear Interaction Vertex Position:**
 - Estimated depth (z-position) where the hadron first undergoes a nuclear interaction in the calorimeter.
- **Number of Non-Empty Cells**
 - Count of calorimeter cells with energy deposit > 0.1 MeV.
- **Total Energy Deposited**
 - Sum of all energy deposited across calorimeter cells in an event.
- **Radius (R)**
 - Average transverse spread of energy deposits around the shower's center.



Boosted Decision Trees (BDT)

- XGBoost is an efficient gradient boosting algorithm that builds an ensemble of decision trees sequentially, where each new tree minimizes the errors of previous ones using gradient descent on a defined loss function.
- It includes regularization to prevent overfitting and uses **3-fold cross-validation** to optimize parameters.
- The model uses the **binary:logistic** objective for binary classification.



Deep Sets Architecture

- Deep Sets: If a function can be expressed in the following form

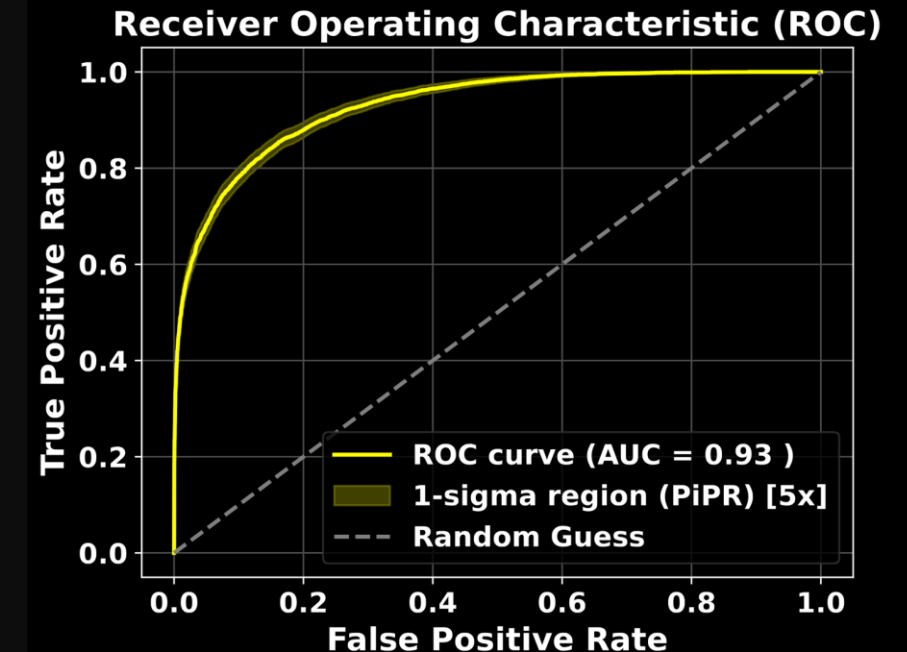
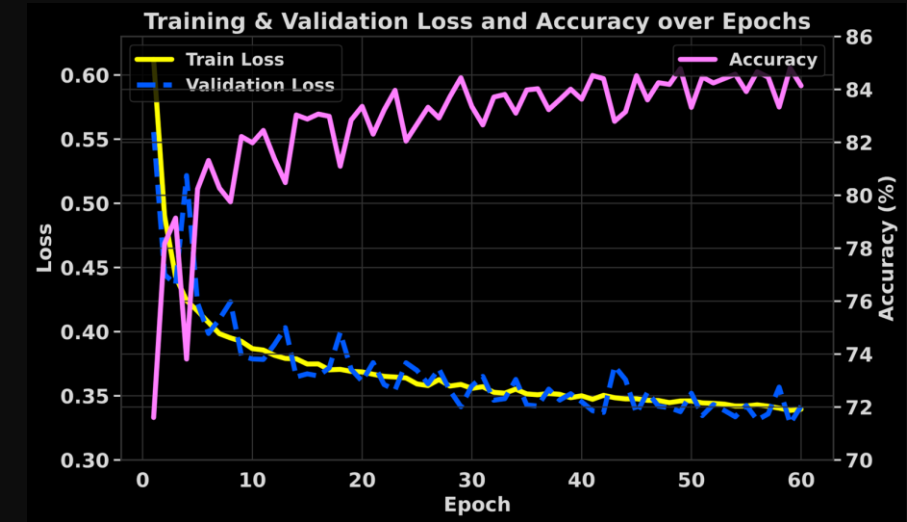
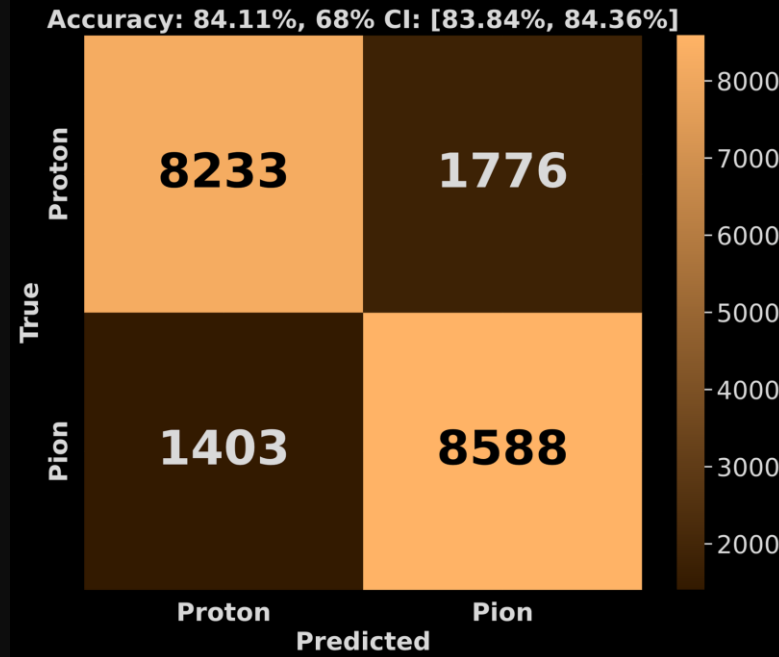
$$f(X) = \rho \left(\sum_{x \in X} \phi(x) \right)$$

- The functions ρ and Φ can be learned by a neural network
- $f(X)$ is a permutation-invariant function since the sum is independent of order of elements in the set X .
- Other similar operations, such as max and mean, can also be applied.

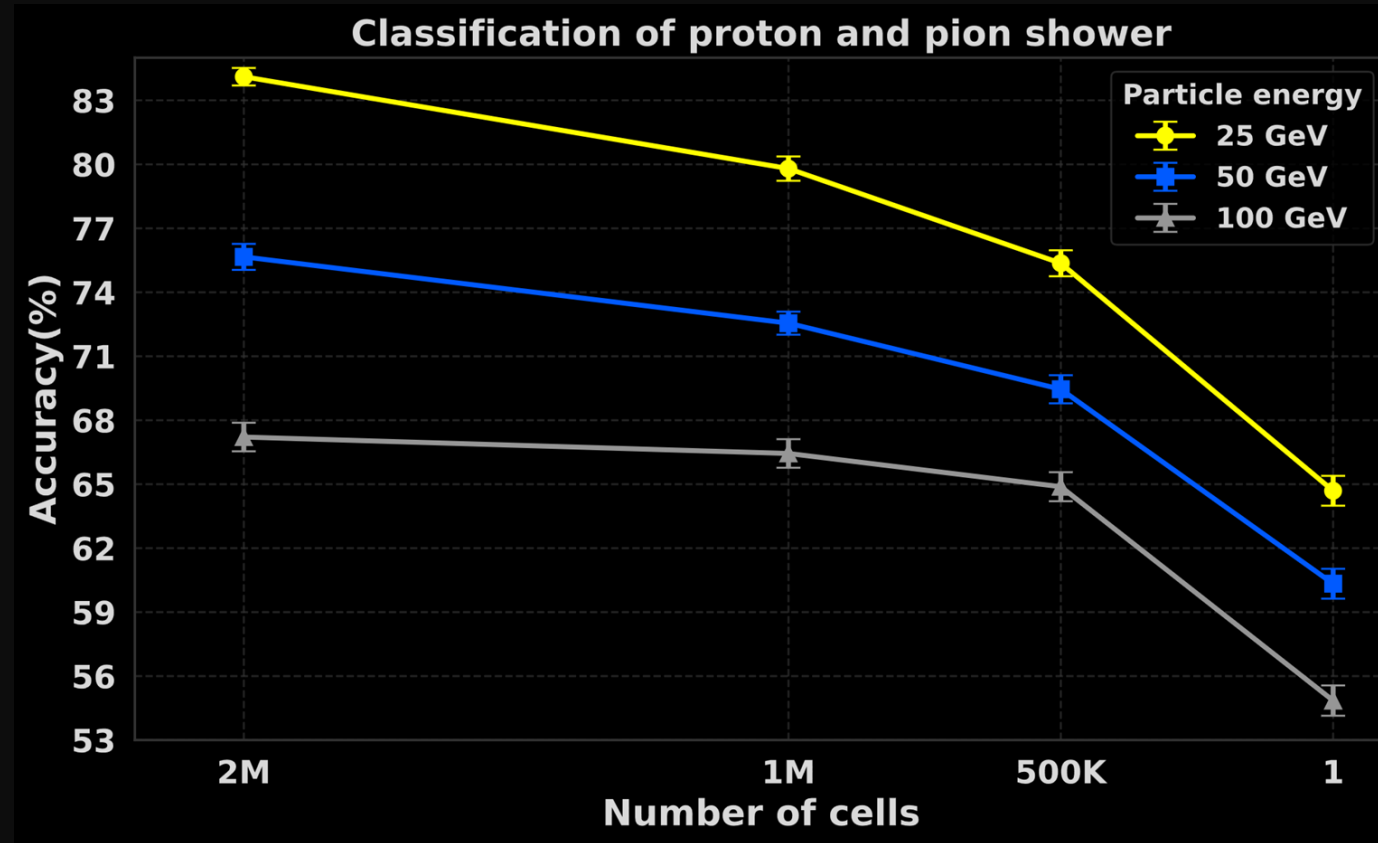
Pooling Function	Permutation-Invariant $f(X)$	Name
Sum	$f(X) = \rho \left(\sum_{x \in X} \phi(x) \right)$	Deep Sets
Average	$f(X) = \rho \left(\frac{1}{ X } \sum_{x \in X} \phi(x) \right)$	Normalized-Deep Sets
Max	$f(X) = \rho \left(\max_{x \in X} (\phi(x)) \right)$	PointNet

Training Results

- Data contains 100K events each for p and π^+ .
- The NN is trained with 60% of events.
- 20% is used for validation and 20% for testing.
- NN model is saved at checkpoint with lowest **validation** loss.
- The uncertainty in accuracy is calculated with **clopper-pearson** method.



Results



Conclusion and Future Work

- Studied the impact of calorimeter granularity on particle identification
- Classification performance is directly proportional to the granularity of calorimeter.
- Impact of granularity is significantly higher at lower energy.

Future Work

- Simulate a more realistic calorimeter with spatial and temporal resolution and test the impact of granularity.
- Explore the spatial temporal features separately.
- Combine the high-level features.
- Extend to multiple particle identification.

Thank you for your attention!

PRODUCT DETAILS

	PbWO ₄	Unit
Density	8.28	g/cm ³
Radiation Length	0.92	cm
Decay Constant	6/30	ns
Wavelength of Emission Peak	440/530	nm
Light yield(of NaI:Tl)	0.5	%
Melting Point	1396	K
Hardness	/	mohs
refractive Index	2.16	/
Hygroscopic	no	/
Cleavage	(101)	/

3.3. Local Variables

The introduction of longitudinal and transverse segmentation in the calorimeter enables the study of the evolution of the energy shower as the particle interacts with the calorimeter. Based on this concept and considering the physical properties of the particles under examination, it is possible to define a set of local variables:

- **First nuclear interaction vertex position:** The position of the first nuclear interaction vertex provides an indirect measure of the probability that a particle will interact with the medium through which it is passing. This probability, represented by the particle's nuclear cross section, depends on the properties of the medium, the energy of the particle, and the particle's identity. Therefore, when the first two factors are held constant, the position of the first interaction vertex becomes a variable sensitive to the particle's identity. To determine this position, the First Nuclear Interaction Vertex Finder is used (see Appendix A).
- **First interaction vertex time:** The instant at which the first nuclear interaction vertex takes place can be defined as the characteristic time of the cell identified as containing that vertex.
- **Speed:** Given the first nuclear interaction vertex position and the first interaction vertex time, the particle speed is defined as the ratio between these two quantities.
- Δt : Given the first interaction vertex Time t_V and the time when 50% of the total deposited energy is exceeded (t_{50}), it is possible to define $\Delta t = t_V - t_{50}$.
- **Fraction of energy deposited after the first vertex:** Referring to the longitudinal segmentation, the calorimeter can be thought as a collection of layers perpendicular to the direction of the primary particle. Hence, the fraction of energy released after the vertex is defined as the energy released in the calorimeter cells located after the layer containing the vertex.
- **Number of non-empty cells before the first vertex layer:** Using the same logic applied to compute the fraction of energy deposited after the first vertex, it is also possible to count the number of non-empty cells in the layers preceding the one containing the vertex.
- **Number of non-empty cells:** The total number of cells for which the deposited energy is greater than 0.1 MeV.
- **Maximum cell energy:** Maximum cell energy refers to the highest total energy deposited in the cells of the calorimeter.
- **Second maximum cell energy:** This variable measures the total energy deposited in the calorimeter cells, representing the second highest value.
- **Total energy close to the first vertex and fraction of energy close to the first vertex:** Once the cell of the primary vertex has been identified, it is possible to define a sphere with radius d (cell unit), centered on the selected cell. The total energy deposited in the cells within this sphere represents the total energy close to the first vertex. Thus it can be used to compute the fraction of the total energy deposited in the calorimeter.
- **Maximum energy deposited close to the first vertex:** Once the cell of the primary vertex has been identified, it is possible to define a sphere with radius d (cell unit),

- **Left and right energy deposition asymmetry:** The impinging position of the primary particle can be considered the center of the reference system, so it is necessary to change the reference system from that of the simulation to the one just described:

$$x^* = x - x_c \quad (6)$$

$$y^* = y - y_c \quad (7)$$

Once the new reference system has been adopted, it is possible to compare the left and right energy deposition. There are two methods: the standard definition (E^{LR}) and the geometrical definition ($\bar{E}^{LR}$). The former is defined as follows

$$E_x^{LR} = \sum_{i=1}^N \text{sgn}(x_i^*) \cdot E_i \quad \text{where } \text{sgn}(x_i^*) = \begin{cases} 1 & \text{if } x_i^* > 0 \\ -1 & \text{if } x_i^* < 0 \\ 0 & \text{if } x_i^* = 0 \end{cases} \quad (8)$$

$$E_y^{LR} = \sum_{i=1}^N \text{sgn}(y_i^*) \cdot E_i \quad \text{where } \text{sgn}(y_i^*) = \begin{cases} 1 & \text{if } y_i^* > 0 \\ -1 & \text{if } y_i^* < 0 \\ 0 & \text{if } y_i^* = 0 \end{cases} \quad (9)$$

$$E^{LR} = \sqrt{(E_x^{LR})^2 + (E_y^{LR})^2} \quad (10)$$

The geometrical definition, on the other hand, is defined as the following:

$$\bar{E}_x^{LR} = \sum_{i=1}^N x_i^* \cdot E_i \quad (11)$$

$$\bar{E}_y^{LR} = \sum_{i=1}^N y_i^* \cdot E_i \quad (12)$$

$$\bar{E}^{LR} = \sqrt{(\bar{E}_x^{LR})^2 + (\bar{E}_y^{LR})^2} \quad (13)$$

- R_E^{cell} : The energy ratio R_E^{cell} is defined as the following

$$R_E^{cell} = \frac{E_{max} - E_{2^{nd}max}}{E_{max} + E_{2^{nd}max}} \quad (14)$$

Here, E_{max} represents the maximum total energy deposited in one cell and $E_{2^{nd}max}$ is the second maximum total energy.

- Δ_E^{cell} : The energy Delta Δ_E^{cell} is the numerator of R_E^{cell} .
- F_E : The energy fraction is defined as the following

$$F_E = \frac{E(\text{within up to } \pm N \text{ cells around } E_{max})}{E(\text{within up to 1 cell around } E_{max})} - 1 \quad (15)$$

Here, N can be tuned and it defines a cube around the cell with the maximum total energy.

BACK-UP

- Reconstructs each particle using the most precise subdetector:
 - Tracking → charged particles
 - ECAL → photons
 - HCAL → neutral hadrons

